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RESEARCH-ARTICLE

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Distillation-enhanced Graph Masked Autoencoders for Bundle Recommendation

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ABSTRACT

Bundle recommendation aims to recommend a bundle of items to users as a whole with user-bundle (U-B) interaction information, and auxiliary user-item (U-I) interaction and bundle-item affiliation information. Recent methods usually use two graph neural networks (GNNs) to model user's bundle preferences separately from the U-B graph (bundle view) and U-I graph (item view). However, by conducting statistical analysis, we find that the auxiliary U-I information is far underexplored due to the following reasons: 1) Loosely combining the predicted results cannot well synthesize the knowledge from both views. 2) The local U-B and U-I collaborative relations might not be consistent, leading to GNN's inaccurate modeling of user's bundle preference from the U-I graph. 3) The U-I interactions are usually modeled equally while the significant ones corresponding to user's bundle preference are less emphasized.

Based on these analyses, we propose a Distillation-enhanced Graph Masked AutoEncoder (DGMAE) for bundle recommendation. Our framework extracts the knowledge of first- and higher-order U-B relations from the U-B graph and injects it into a well-designed graph masked autoencoder (student model). The student model is built with two key designs to jointly capture significant local and global U-I relations from the U-I graph. In specific, we design a transformer-enhanced GNN encoder for global relation learning, which increases the model's representational power of depicting user's bundle preferences. Meanwhile, an adaptive edge masking strategy and reconstruction target are designed on the significant U-I edges to guide the student model to identify the potential ones suggesting user's bundle preferences. Extensive experiments on benchmark datasets show the significant improvements of DGMAE over the SOTA methods.

CCS CONCEPTS

• Information systems → Collaborative filtering.

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1 INTRODUCTION

Recommender system (RS) [10, 28] has been an effective tool to alleviate the information overload issue [29, 35]. Generally, conventional RS aims to recommend individual items to users. But in many E-commerce applications, such as fashion shopping sites, item-bundling is a widely-adopted marketing strategy to avoid users' monotonous choices, thus improving user's experience. From the aspect of platforms, the bundled sales strategy can increase sales revenue by adding items that are seldom purchased separately to the bundle. To this end, the problem of bundle recommendation [36] is proposed to recommend a list of items to users as a whole. In addition to the user-bundle (U-B) interaction information, the auxiliary user-item (U-I) interaction and bundle-item (B-I) affiliation information are available for more fine-grained modeling on user's bundle preferences from the item level.

Existing bundle recommendation studies [3, 4, 6, 21] mainly fall into two classes: One class of methods [4, 6] simultaneously model U-B and U-I interactions in a holistic manner. For example, BundleNet [6] merges the U-B and U-I interactions into a tripartite graph and employs graph neural networks (GNNs) to aggregate the local information into user/bundle representations. However, these methods cannot explicitly model the relationship between users, items and bundles [3] and fail to model the cooperative association between the two types of interactions [21]. Another class of methods [3, 21] move them forward by separately modeling the U-B and U-I interactions in two views. For instance, BGCN [3] designs two GNNs to learn user's bundle preference from the U-B graph (bundle view) and U-I graph (item view) and combines the predicted results from two views. CrossCBR [21] follows the framework of BGCN and introduces a contrastive task to maximize the agreements of user/bundle representations from two views.

Despite the success, we argue that the auxiliary U-I information still has far not reached its potential due to the following issues:

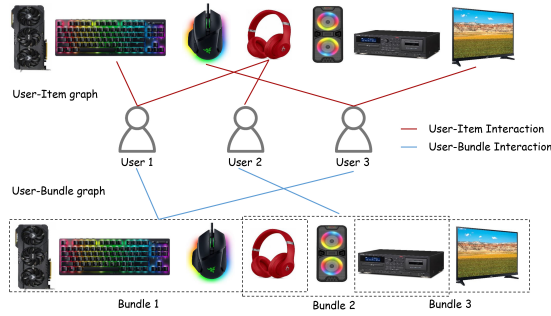


Figure 1: An example of user-bundle and user-item graphs.

- The simple combination of predicted results adopted by existing works [3, 21] cannot effectively integrate the knowledge from two views. Although CrossCBR alleviates this issue by adding a contrastive task, it fails to transfer the knowledge of higher-order relationships among users and bundles between two views.
- Users with similar item preferences might not also have similar bundle preferences. As shown by the example in Figure 1, Users 1 and 2 both have interacted with a headset but tend to choose different bundles due to their different preferences on games and music. This phenomenon leads to the inconsistency of local U-B and U-I collaborative relations, further implying the limitations of GNNs to learn user's bundle preferences on the U-I graph.
- The U-I interactions are not equally significant for predicting U-B interactions whereas equally treated by existing methods. As shown in Figure 1, User 3 has interacted with a television but rejects the bundle containing it. By contrast, he/she selects Bundle 1 while interacting with the mouse in it. Thus we assume that the U-I relations corresponding to user's bundle preference play a more important role in bundle recommendation.

To validate the proposed assumptions, we conduct statistical analyses based on which we propose a Distillation-enhanced Graph Masked AutoEncoder (DGMAE) to tackle the issues. Specifically, DGMAE pretrains a teacher model to extract the knowledge of first- and higher-order relations among users and bundles from the U-B graph. Then it injects the knowledge into a carefully designed graph masked autoencoder (student model) on the U-I graph through a distillation object. There are two key designs in the student model: 1) DGMAE adds a transformer subnetwork on top of a GNN encoder to jointly capture local and global U-I collaborative relations. For consideration of efficiency, we introduce two sets of virtual nodes for users and items in the transformer layer and perform a three-stage global propagation among user/item nodes and virtual nodes. 2) DGMAE adopts an adaptive masking strategy on the significant edges in U-I graph and encodes the partially-masked graph. By reconstructing these masked edges, the encoder is able to identify the potential U-I interactions associated with user's bundle preferences. In addition, these two designs increase DGMAE's robustness to the sparsity issue of U-B interactions [3] and the noise issue of U-I interactions [31]. Consequently, DGMAE integrates the knowledge from both views and explores the U-I interaction information in a more accurate and fine-grained way.

Our contributions can be mainly summarized as:

- We conduct statistical analyses and empirical experiments to provide guidance about how to leverage U-I interaction information in a more effective way for bundle recommendation.

- We propose a knowledge distillation framework to transfer higher-order relations among users and bundles from the bundle view to the item view for more effective knowledge integration.
- We design a graph masked autoencoder to dive deeper into the U-I interaction information for more fine-grained learning on user's bundle preferences.
- Extensive experiments on benchmark datasets show that DGMAE outperforms the SOTA methods by a large margin.

2 RELATED WORK

2.1 Bundle Recommendation

Bundle recommendation aims to recommend a bundle of items to users for consumption as a whole. Early works such as LIRE [20] and EFM [1] simultaneously took into consideration users' previous interactions with item lists and individual items based on the BPR framework. After them, DAM [4] employed an attention network to model U-B interactions and U-I interactions in a multi-task manner. In recent years, the success of GNN in modeling the graph data also inspired its application in bundle recommendation. For example, BundleNet [6] performed message passing on the user-item-bundle tripartite interaction graph. BGCN [3] outperformed BundleNet by decomposing users' preference into item view and bundle view and used two GCNs to model the two types of preferences, respectively. Based on BGCN, CrossCBR further [21] modeled the cooperative association between the two views through contrastive learning and achieved the SOTA performance. Different from these works, we integrate the knowledge from two views through a knowledge distillation framework and propose a graph masked autoencoder to model the user's bundle preferences from the item view more accurately.

2.2 Knowledge Distillation

Knowledge Distillation (KD) [11] was proposed for model compression by transferring knowledge from a large teacher model to a small light-weight student model and widely used in the computer vision (CV) area [2, 19, 23]. The hidden properties behind the training set are extracted from the teacher and injected into the learned student, thus improving the performance of the student model. Inspired by its success in CV, some recent studies have introduced KD to RS for efficiency improvement of the existing RS systems [13, 17, 18]. Different from them, we transfer the first- and higher-order relations among users and bundles from bundle view to item view for better knowledge integration.

2.3 Masked Autoencoders

Traditional autoencoders [15, 16] usually take complete data instances as input and target to reconstruct the entire input data. Different from them, masked autoencoding randomly masks a proportion of input data, encodes the partially-masked data, and aims to reconstruct the masked tokens. By setting a more challenging and meaningful self-supervised task, masked autoencoders have recently achieved great success in modeling graphs [12], images [9], and texts [7]. In this work, DGMAE designs an adaptive masking strategy and a transformer-enhanced encoder, which not only improves the expressive power in learning user representations but also effectively alleviates the sparsity issue of U-B interactions in bundle recommendation.

Table 1: Statistics of bundle recommendation datasets.

Datasets	Youshu	NetEase	iFashion
#Users M	8,039	18,528	53,897
#Items O	32,770	123,628	27,694
#Bundles N	4,771	22,864	42,563
# Avg. I/B	37.03	77.80	3.86
$ \mathcal{E}_{UI} $	138,515	1,128,065	2,290,645
$ \mathcal{E}_{UB} $	51,377	302,303	1,679,708
$ S_B $	1,621,817	8,714,602	222,672,479
$ S_I $	6,676,502	78,124,204	508,792,309
$ S_B \cap S_I $	337,081	2,600,826	106,700,889
c-score	0.0812	0.0599	0.2917
$ \mathcal{E}_I $	18,672	141,783	754,723
$ \mathcal{E}_D $	119,843	986,282	1,535,922

3 PRELIMINARIES

3.1 Problem Definition

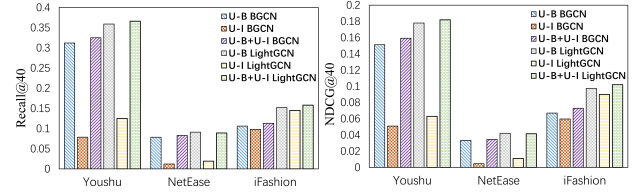
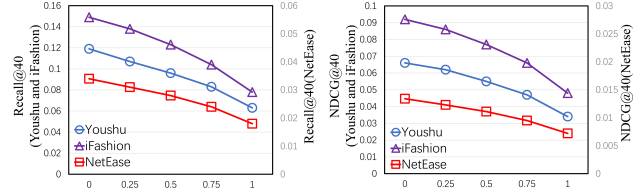
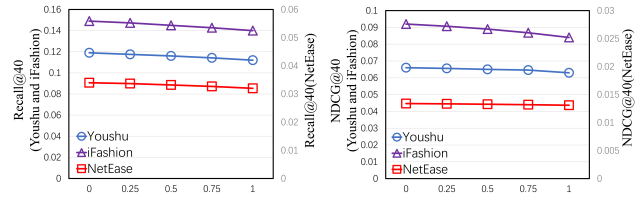
In the task of bundle recommendation, we denote $\mathcal{U}$ ($|\mathcal{U}| = M$) as the user set, $\mathcal{B}$ ($|\mathcal{B}| = N$) as the bundle set and $\mathcal{V}$ ($|\mathcal{V}| = O$) as the item set. The user-bundle interactions, user-item interactions, and bundle-item affiliations are represented as $X_{M \times N} = \{x_{ub} \mid u \in \mathcal{U}, b \in \mathcal{B}\}$, $Y_{M \times O} = \{y_{uv} \mid u \in \mathcal{U}, v \in \mathcal{V}\}$ and $Z_{N \times O} = \{z_{bv} \mid b \in \mathcal{B}, v \in \mathcal{V}\}$ with a binary value at each entry, respectively. The value 1 denotes an observed interaction between the user-bundle or user-item pair, or bundle-item containment. It is notable that X and Y are separately generated, allowing users to interact with both bundles and individual items. The task is to predict the unseen user-bundle interactions from the historical $\{X, Y, Z\}$.

3.2 Data Analysis

In order to figure out how to leverage the U-I interaction information in a more effective way, we perform statistical analyses on three widely used bundle recommendation datasets, including *Youshu* [4], *NetEase* [1], and *iFashion* [5]. Specifically, we begin with performing ablation studies on the framework of BGCN [3] and CrossCBR [21] to explore the effect of item view. Then we design metrics and extensive experiments to explore the in-depth reasons behind the results. In summary, we aim to answer the following three research questions:

- RQ1: How do the bundle view and item view contribute to the overall performance of bundle recommendation? Can they complement each other well?
- RQ2: Are the U-I and U-B collaborative relations consistent with each other? In other words, do users with similar preferences on individual items also have similar preferences on bundles?
- RQ3: Are the edges in the U-I graph equally important? If not, which interactions have a greater impact on predicting user's preference on bundles?

3.2.1 RQ1. To validate the effect of bundle and item views and their complementarity, we conduct experiments on the three datasets and compare the results among the U-B GNN, U-I GNN, and the combination of two GNNs (U-B+U-I GNN). Specifically, we employ BGCN [3] and LightGCN [10] (GNN used in CrossCBR [21]) as the representative methods and optimize the same BPR loss [24] in the two views, respectively. In the U-I GNNs, we follow [3, 21] to obtain bundle representations through average pooling on the item

**Figure 2: Performances of the two GCNs on the user-bundle and user-item views**(a) Edge removal in $\mathcal{E}_I$ with different proportions.(b) Edge removal in $\mathcal{E}_D$ with the same removal number in $\mathcal{E}_I$.**Figure 3: Performances of LightGCN on the modified user-item graphs.**

representations. The predicted scores of the two views are summed as the final prediction of the combination model. For all experiments, we randomly divide the datasets into training, validation, and testing sets with a ratio of 7:1:2. The results are presented in Fig. 2 and the observations are as follows.

1) Except for comparable results on iFashion, the performances of U-I GNNs are far inferior to U-B GNNs (only about 30% performance). We leave the detailed analysis of this result to RQ2 and RQ3. 2) The improvements of the U-B+U-I GNNs to the U-B GNNs on all the datasets are quite limited (<5%). On NetEase, directly involving the information from the item view even slightly hurts the performance of LightGCN, which highlights a desire for a more effective knowledge integration method.

3.2.2 RQ2. In order to explore the in-depth reason for GNN's poor performance on the U-I graph, we first briefly review the GNN principle from the graph signal processing perspective. Previous works [25, 30] have shown that node embeddings consist of low-frequency true features and high-frequency noise. GNNs iteratively aggregating neighbourhood information is equivalent to performing low-pass filtering to denoise node features. It is reasonable to perform such operators on the U-B graph since the data to be predicted has the same distribution with the observed data. However, whether users with similar item preferences also have similar bundle preferences has not been analyzed before. If the answer is not, smoothing user embedding according to U-I interactions will limit model's expressive power of depicting user's bundle preferences.

In response to this question, we view the U-B or U-I interactions as evidence to reflect user's bundle/item preferences. In particular,

we first define two sets of user pairs: $S_B = \{(u_1, u_2) | \exists v \in \mathcal{N}_X(u_1) \cap \mathcal{N}_X(u_2)\}$ (two users interact with at least one same bundle) and $S_I = \{(u_1, u_2) | \exists v \in \mathcal{N}_Y(u_1) \cap \mathcal{N}_Y(u_2)\}$ (two users interact with at least one same item) to denote the user pairs with similar bundle preferences and item preferences, respectively. Then we design a metric *c-score* to measure the consistency between U-B and U-I collaborative relations, which is defined as follows:

$$c\text{-score} = \frac{2|S_B \cap S_I|}{|S_I| + |S_B|}. \quad (1)$$

The design of *c-score* is inspired by the famous *F1-score*, balancing the hit ratio of user pairs in S_I and the coverage ratio of user pairs in S_B . Generally speaking, if the *c-score* in a dataset is high, we can draw the conclusion that users with similar item preferences also have similar bundle preferences.

The results are shown in Table 1. We observe that the *c-scores* of Youshu and NetEase are quite low (<0.1), meaning that there's quite limited consistency between the U-B and U-I collaborative relations. Note that this result is consistent with U-I GNNs' poor performance on these two datasets and further points out a challenge for increasing the model's expressive power and jointly capturing local and global U-I collaborative relations.

3.2.3 RQ3. Another fact in existing employed GNNs is that it iteratively aggregates the neighbourhood node embeddings without differentiating the importance of edges. However, users are more cautious when making decisions for buying a bundle than buying an item due to the higher costs of bundles in practice [3]. It is common for users to select several items whereas rejecting the bundle containing them, bringing huge challenges to differentiate the effective interactions from the noisy ones in the U-I graph.

To discriminate the U-I interactions, we construct another U-I graph $Y' = XZ$ where we suppose that users interact with each item in the bundles they have interacted with. Then we divide the edges in Y into two sets: $\mathcal{E}_I = \{(u, i) | Y_{ui} = 1, Y'_{ui} > 0\}$ denoting the intersection edge set between Y and Y' , and $\mathcal{E}_D = \{(u, i) | Y_{ui} = 1, Y'_{ui} = 0\}$ denoting the difference edge set between Y and Y' . We assume that the edges in $\mathcal{E}_I$ contribute more in predicting user's bundle preference since they directly indicate the items that users interact with bundles for. For simplicity, we name these edges as *effective interactions* and others as *background interactions* in the rest of paper. To validate this assumption, we corrupt the U-I graph by randomly dropping out the edges in $\mathcal{E}_I$ with different ratios and removing the same corresponding amount of edges in $\mathcal{E}_D$ for comparison. We employ LightGCN on the modified U-I graphs and show the results in Fig. 3.

The observations are as follows. 1) GNNs are robust to the removal of edges in $\mathcal{E}_D$ on all the datasets, which shows that most edges in $\mathcal{E}_D$ cannot provide useful information about user's preference on bundles. 2) By contrast, GNN's performance shows an apparent decrease with the gradual removal of the edges in $\mathcal{E}_I$, indicating the importance of edges in $\mathcal{E}_I$. Therefore, we conclude that another key challenge is to recalibrate the U-I relations by preserving the effective relations while denoising the useless ones.

4 METHODOLOGY

This section gives a detailed introduction to our proposed DGMAE framework and its overview is shown in Figure 4. We first present a

knowledge distillation framework to effectively synthesize knowledge from the bundle and item views (RQ1). Afterwards, we will illustrate the architecture of our student model which can jointly capture local and global user-item collaborative relations (RQ2) and identify the potential effective interactions (RQ3). Finally, we will discuss the model computation complexity.

4.1 The Knowledge Distillation Framework

In the bundle recommendation task, the additional U-I interaction and B-I affiliation information enable us to decompose the bundle into finer-grained items and infer the reason for user's interaction with bundles at the item level. Thus we propose to use the knowledge distillation (KD) technique to transfer the knowledge from bundle view to item view for more fine-grained learning of user's bundle preferences. In our KD framework, we employ LightGCN on the U-B graph X as the teacher model due to its good performance as presented before. Specifically, the embedding propagation process in the l -th layer of teacher model is represented as:

$$e_u^{T(l)} = \sum_{b \in \mathcal{N}_X(u)} \frac{e_b^{T(l-1)}}{\sqrt{|\mathcal{N}_X(u)|} \sqrt{|\mathcal{N}_X(b)|}}, \quad (2)$$

where $e_u^{T(l)}$ denotes the user embedding learned by the teacher model in the l -th layer. $e_u^{T(0)}, e_b^{T(0)} \in \mathcal{R}^d$ are randomly initialized parameters. Similarly, we can obtain the message passing from users to bundles. Then a layer aggregation operation is performed to gather the information from L_T layers:

$$e_u^T = \sum_{l=0}^{L_T} e_u^{T(l)}, \quad e_b^T = \sum_{l=0}^{L_T} e_b^{T(l)}, \quad (3)$$

where e_u^T and e_b^T are the learned user and bundle embeddings by the teacher. Finally, we calculate the user-bundle predictions utilizing the inner-product and pretrain the teacher model by optimizing the BPR loss [24]:

$$\mathcal{L}^{BPR} = \sum_{(u,b,b') \in \mathcal{T}} -\ln \sigma(y_{u,b}^T - y_{u,b'}^T), \quad y_{u,b}^T = e_u^{T\top} e_b^T, \quad (4)$$

where $\mathcal{T} = \{(u, b, b') | u \in \mathcal{U}, b, b' \in \mathcal{B}, X_{ub} = 1, X_{ub'} = 0\}$.

After deriving the pretrained teacher model, an important question is what knowledge should be transferred to the student model. Previous works [17, 26] usually train the student to mimic the soft predictions made by the teacher. However, we argue that simply preserving the U-B relations will lose the higher-order information in the U-B graph. Thus we also align the output relations between users/bundles of the teacher model to the student model. Specifically, we use $f_{STU;\theta}$ to denote the student model parameterized by θ , which will be elaborated later. Let $e_u^S = f_{STU;\theta}(u)$ and $e_b^S = f_{STU;\theta}(b)$ denote the embeddings of user u and bundle b output by the student model. The distillation loss is defined as:

$$\mathcal{L}^{KD} = \sum_{(u,b)} \mathcal{L}^B(y_{u,b}^T, y_{u,b}^S) + \sum_{(u,u')} \mathcal{L}^B(y_{u,u'}^T, y_{u,u'}^S) + \sum_{(b,b')} \mathcal{L}^B(y_{b,b'}^T, y_{b,b'}^S) \quad (5)$$

where (u, u') , (b, b') , and (u, b) are sampled from a batch in experiments. $y_{u,v} = e_u^{S\top} e_v^S$ denotes the similarity score calculated by the student model and $\mathcal{L}^B$ denotes the binary cross-entropy loss.

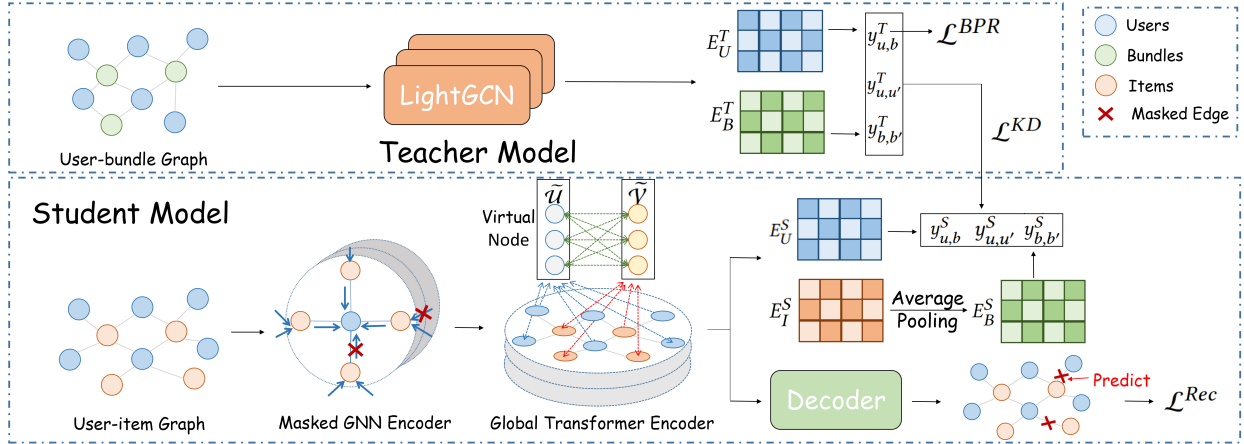


Figure 4: Overview of DGMAE.

4.2 The Student Model

In this part, we design a self-supervised graph masked autoencoder as the student model to tackle the challenges in RQ2 and RQ3. There are four components in the student model: masked GNN encoder, global transformer encoder, decoder, and the reconstruction target, which will be introduced in detail as below.

4.2.1 Masked GNN Encoder. According to the result of RQ3, the U-I interactions that reflect user's bundle preference play a more significant role compared to the background interactions. Thus the key object of encoder is to leverage existing effective U-I interactions to infer the unobserved ones. Taking from the essence of recent proposed masked autoencoder [9] in computer vision, we propose to mask a proportion of edges in $\mathcal{E}_I$ and perform embedding propagation only on the partially-masked U-I graph. Guided by the reconstruction object on the masked edges, the encoder can learn to recover the significant missing information from the masked graph and further infer the latent effective interactions. Based on this insight, we then describe our edge masking strategy.

For each edge $(u, i) \in \mathcal{E}_I$, we randomly mask it with probability $p_{ui} \in (0, 1)$ to divide $\mathcal{E}_I$ into the masked set $\mathcal{E}_I^M$ and reserved set $\mathcal{E}_I^R$. A direct masking strategy is randomly masking a certain proportion of edges by setting p_{ui} as a fixed number. However, the sparse U-B interactions lead to the fact that the edges in $\mathcal{E}_I$ only take a small proportion of all edges (e.g. 13.5% in YouShu and 12.6% in NetEase). Thus to alleviate the sparsity issue, we propose to pay more attention to users/items with sparse effective interactions. Specifically, we first calculate the sparsity score s for each edge based on the effective interaction degrees of user and item:

$$s_{ui} = \frac{\deg(u) + \deg(i)}{2}, \quad \deg(u) = |\{(u, i) \in \mathcal{E}_I\}|. \quad (6)$$

Next, we calculate the probability of each edge based on its sparsity score. Since the degrees of nodes may vary across orders of magnitude [22], we set $w_{ui} = \log s_{ui}$ to alleviate the impact of nodes with heavily dense connections. Then we normalize the sparsity values to obtain the probability:

$$p_{uv}^e = p_l + \frac{w_{\max} - w_{uv}}{w_{\max} - w_{\min}} \cdot (p_u - p_l), \quad (7)$$

where $p_l < p_u < 1$ are the hyperparameters that control the lower bound and upper bound of masking probability, respectively. $w_{\max}$

and $w_{\min}$ denote the maximum and minimum values of w_{ui} . Generally, an edge with low sparsity score has a higher probability to be masked, making the reconstruction more challenging and increasing model's robustness to the sparsity issue.

After deriving the masked and reserved edge subset $\mathcal{E}_I^M$ and $\mathcal{E}_I^R$, we combine the edges in $\mathcal{E}_I^R$ and $\mathcal{E}_D$ to construct the partially-masked U-I graph $\tilde{Y}$, i.e., $\tilde{Y}_{ui} = 1, (u, i) \in \mathcal{E}_I^R \cup \mathcal{E}_D$. Then we conduct embedding propagation over $\tilde{Y}$ via LightGCN where the propagation in the l -th layer is denoted as:

$$e_u^{(l)} = \sum_{v \in \tilde{N}_{\tilde{Y}}(u)} \frac{e_v^{(l-1)}}{\sqrt{|\tilde{N}_{\tilde{Y}}(u)|} \sqrt{|\tilde{N}_{\tilde{Y}}(v)|}}, \quad (8)$$

where $e_u^{(l)}$ denotes the learned user embedding in the l -th layer. $e_u^{(0)}, e_b^{(0)} \in \mathcal{R}^d$ are randomly initialized parameters and we share the parameters of $e_u^{(0)}$ and $e_u^{T(0)}$ for space efficiency consideration. We then stack L_S layers to form the final representations for users/items learned by the local encoder:

$$e_u = \sum_{l=0}^{L_S} e_u^{(l)}, \quad e_v = \sum_{l=0}^{L_S} e_v^{(l)}. \quad (9)$$

4.2.2 Global Transformer Encoder. According to the result of RQ2, simply using GNN to smooth user/item representations with U-I interactions might lead to unsatisfactory results due to the inconsistency between U-B and U-I collaborative relations. To address this limitation, we further enhance the encoder with a transformer sub-network which increases the model's expressive power and allows long-range embedding propagation for global relation learning.

It is notable that direct information exchange between all the U-I pairs leads to a prohibitive computational cost of $O(MO)$. Thus to balance efficiency and effectiveness, we introduce two virtual node (Vnode) sets $\tilde{\mathcal{U}}$ and $\tilde{\mathcal{V}}$ of size K for users and items as the medium. These two Vnode sets enable information exchange through a three-stage information propagation among nodes and Vnodes where the weights of connections are optimized to reflect the implicit dependencies among users and items. For simplicity, we mainly discuss the propagation process of users.

Node-to-Vnode propagation. We first perform information propagation from the user nodes to the user-side Vnodes. Specifically, for a Vnode $\tilde{u} \in \tilde{U}$, the propagation function is built upon an attention network [27]:

$$\tilde{\mathbf{p}}_{\tilde{u}} = \|\mathbf{p}_{\tilde{u}}^H\|_{h=1}^H; \quad \tilde{\mathbf{p}}_{\tilde{u}}^h = \sum_{u \in \mathcal{U}} \alpha_{u,u}^h \tilde{\mathbf{e}}_{u, \frac{(h-1)d}{H} : \frac{hd}{H}}; \quad (10)$$

$$\alpha_{u,u}^h = \text{softmax}_{u' \in \mathcal{U}} \left(a_{u,u'}^h \right); \quad a_{u,u}^h = \left(\mathbf{W}_{Q_1}^h \tilde{\mathbf{p}}_{\tilde{u}} \right)^\top \left(\mathbf{W}_{Q_1}^h \tilde{\mathbf{e}}_u \right) / \sqrt{d},$$

where $\tilde{\mathbf{p}}_{\tilde{u}} \in \mathcal{R}^d$ denotes the embedding of Vnode $\tilde{u}$ and is initialized as a parameter vector. It is obtained by the concatenation of H head-specific embeddings $\tilde{\mathbf{p}}_{\tilde{u}}^h \in \mathcal{R}^{d/H}$. $\tilde{\mathbf{e}}_u$ is the embedding of user u and derived from the local encoding $\mathbf{e}_u$ at the beginning. $\alpha_{u,u}^h \in \mathcal{R}$ denotes the h -head-specific quantitative weight for user u and Vnode $\tilde{u}$, and subscript $\frac{(h-1)d}{H} : \frac{hd}{H}$ denotes the h -th slice of the vector. To alleviate the overfitting effect, we optimize our propagation function from the following perspectives: 1) We slice the d -dimensional $\mathbf{e}_u$ to H chunks as head-specific feature vectors instead of performing the heavy value transformation. 2) We enable the matrix transformation sharing among the query and key calculations with the projection matrix $\mathbf{W}_{Q_1}^h$.

Vnode-to-Vnode propagation. To further distill the collaborative user-item relations from a global perspective, we perform information propagation between the user-side and item-side Vnodes, which is defined as follows:

$$\hat{\mathbf{p}}_{\tilde{u}} = \|\mathbf{p}_{\tilde{u}}^H\|_{h=1}^H; \quad \hat{\mathbf{p}}_{\tilde{u}}^h = \sum_{\tilde{v} \in \tilde{\mathcal{V}}} \alpha_{\tilde{u},\tilde{v}}^h \tilde{\mathbf{p}}_{\tilde{v}, \frac{(h-1)d}{H} : \frac{hd}{H}}; \quad (11)$$

$$\alpha_{\tilde{u},\tilde{v}}^h = \text{softmax}_{\tilde{v}' \in \tilde{\mathcal{V}}} \left(a_{\tilde{u},\tilde{v}'}^h \right); \quad a_{\tilde{u},\tilde{v}}^h = \left(\mathbf{W}_{Q_2}^h \tilde{\mathbf{p}}_{\tilde{u}} \right)^\top \left(\mathbf{W}_{Q_2}^h \tilde{\mathbf{p}}_{\tilde{v}} \right) / \sqrt{d},$$

where $\tilde{\mathbf{p}}_{\tilde{u}}$ and $\hat{\mathbf{p}}_{\tilde{u}}$ denote the embeddings for Vnode $\tilde{u}$ before and after Vnode-level propagation, respectively.

Vnode-to-node propagation. Finally, we propagate information from the user-side Vnodes back to the user nodes through a similar but reverse process to the node-to-Vnode propagation:

$$\hat{\mathbf{e}}_u = \|\mathbf{e}_u^H\|_{h=1}^H; \quad \hat{\mathbf{e}}_u^h = \sum_{\tilde{u} \in \tilde{\mathcal{U}}} \alpha_{u,\tilde{u}}^h \hat{\mathbf{p}}_{\tilde{u}, \frac{(h-1)d}{H} : \frac{hd}{H}}; \quad (12)$$

$$\alpha_{u,\tilde{u}}^h = \text{softmax}_{\tilde{u}' \in \tilde{\mathcal{U}}} \left(a_{u,\tilde{u}'}^h \right); \quad a_{u,\tilde{u}}^h = \left(\mathbf{W}_{Q_3}^h \hat{\mathbf{p}}_{\tilde{u}} \right)^\top \left(\mathbf{W}_{Q_3}^h \tilde{\mathbf{e}}_u \right) / \sqrt{d},$$

where $\hat{\mathbf{e}}_u$ denotes the refined embedding of user u after the three-stage propagation.

The three-stage propagation forms a global transformer encoding layer (denoted as Glob-Trans), capturing the prominent node-wise relations. To further propagate the encoded global collaborative relations, we stack multiple transformer layers and iteratively learn the long-range user-item dependencies. In form, taking the embedding matrix $\hat{\mathbf{E}}^{(l-1)} \in \mathcal{R}^{M \times d}$ or $\mathcal{R}^{O \times d}$ as input, we apply a residual connection to incorporate the local and global encodings and facilitate gradient propagation:

$$\hat{\mathbf{E}}^l = \text{Glob-Trans}(\hat{\mathbf{E}}^{(l-1)}) + \hat{\mathbf{E}}^{(l-1)}. \quad (13)$$

We take the user embedding $\hat{\mathbf{e}}_u^L$ in the last layer L as the user encoding by student model, i.e., $\mathbf{e}_u^S = \hat{\mathbf{e}}_u^L$ and obtain the item embedding $\mathbf{e}_v^S$ in a similar way. Following [21], DGMAE utilizes an average

pooling to obtain the bundle embeddings:

$$\mathbf{e}_b^S = \frac{1}{|\mathcal{N}_Z(b)|} \sum_{v \in \mathcal{N}_Z(b)} \mathbf{e}_v^S. \quad (14)$$

4.2.3 Decoder and Reconstruction Target. Receiving the latent user/item embeddings from the encoder, the decoder models the correlation between user u and item v through a weight function:

$$\phi(u, v) = \omega \left(\mathbf{w} \odot \mathbf{e}_u^S, \mathbf{w} \odot \mathbf{e}_v^S \right), \quad (15)$$

where ω is the cosine similarity estimation based on the trainable vector $\mathbf{w} \in \mathcal{R}^d$ and $\odot$ denotes the element-wise product. Based on the predicted correlations, the model recovers the masked edges to reconstruct the original U-I graph structure. Specifically, the reconstruction loss is defined as follows:

$$\mathcal{L}^{Rec} = - \sum_{(u,v) \in \mathcal{E}_I^M} \log \frac{\exp \{ \phi(u, v) \}}{\sum_{v' \in \mathcal{V}} \exp \{ \phi(u, v') \} + \sum_{u' \in \mathcal{U}} \exp \{ \phi(u', v) \}}, \quad (16)$$

which emphasizes the correlations of masked edges compared to others. In conjunction with the masking strategy, this object makes DGMAE more robust to the network noise and guides the entire encoder to identify the potential effective U-I interactions.

4.2.4 Model Optimization. We train the student model by jointly optimizing the knowledge distillation loss and self-supervised reconstruction loss:

$$\mathcal{L} = \mathcal{L}^{KD} + \lambda \mathcal{L}^{Rec}, \quad (17)$$

where λ is the hyperparameter to balance the two losses. In experiments, we adopt negative sampling [32] to accelerate the optimization of $\mathcal{L}^{Rec}$ since directly calculating the summation is computationally prohibitive.

4.3 Model Complexity Analysis

In terms of space complexity, the extra parameters involved in the global encoder and decoder of student model are $O(K \times d + d^2)$. The time complexity of DGMAE comes from the following aspects: 1) The complexity of teacher model is $O(|\mathcal{E}_{UB}| \times d)$. 2) In the student model, the local encoder costs $O(|\mathcal{E}_{UI}| \times d)$ while the complexities of the three-stage propagation in the global encoder are $O((M+O) \times K \times d)$, $O(K^2 \times d)$, and $O((M+O) \times K \times d)$, respectively. 3) Since we sample the user-user, bundle-bundle, user-bundle and user-item pairs within a batch, calculating the KD loss and the reconstruction loss both cost $O(B^2 \times d)$ where B denotes the batch size. By merging the above costs, the overall complexity of our model is $O((|\mathcal{E}_{UB}| + |\mathcal{E}_{UI}| + (M+O)K + B^2)d)$. Since the typical value of virtual node number K is much smaller than the node number M and O , the complexity of global encoder $O((M+O) \times K \times d)$ is close to that of GNN encoder $O(|\mathcal{E}_{UI}| \times d)$. Thus both the space and time complexities of our model are comparable to those of Cross-CBR [21], which are $O((M+N+O)d)$ and $O((|\mathcal{E}_{UB}| + |\mathcal{E}_{UI}| + B^2)d)$.

5 EXPERIMENTS

Our experiments aim to answer the following research questions:

- **RQ1:** How does DGMAE compare to the state-of-the-art methods in the bundle recommendation task?
- **RQ2:** How is the robustness of DGMAE to the sparsity and noise issues compared to the existing methods?

Table 2: Performance comparison between DGMAE and the baseline methods. The best and second-best results are highlighted with boldface and underlined. The improvements achieved by DGMAE are significant (p -value $\ll 0.05$).

Model	Youshu				NetEase				iFashion			
	Rec@20	NDCG@20	Rec@40	NDCG@40	Rec@20	NDCG@20	Rec@40	NDCG@40	Rec@20	NDCG@20	Rec@40	NDCG@40
MFBPR	0.1959	0.1117	0.2735	0.1320	0.0355	0.0181	0.0600	0.0246	0.0752	0.0542	0.1162	0.0687
DAM	0.2082	0.1198	0.2890	0.1418	0.0411	0.0210	0.0690	0.0281	0.0629	0.0450	0.0995	0.0579
BundleNet	0.1895	0.1125	0.2675	0.1335	0.0391	0.0201	0.0661	0.0271	0.0626	0.0447	0.0986	0.0574
BGCN	0.2347	0.1345	0.3248	0.1593	0.0491	0.0258	0.0829	0.0346	0.0733	0.0531	0.1128	0.0671
LightGCN	0.2596	0.1467	0.3593	0.1784	0.0563	0.0292	0.0915	0.0384	0.1018	0.0744	0.1523	0.0975
MIDGN	0.2682	0.1527	0.3712	0.1808	0.0678	0.0343	0.1085	0.0451	0.1106	0.0827	0.1628	0.1035
CrossCBR	<u>0.2813</u>	<u>0.1668</u>	<u>0.3785</u>	<u>0.1938</u>	<u>0.0842</u>	<u>0.0457</u>	<u>0.1264</u>	<u>0.0569</u>	<u>0.1173</u>	<u>0.0895</u>	<u>0.1699</u>	<u>0.1080</u>
DGMAE	0.3245	0.1918	0.4323	0.2231	0.0945	0.0512	0.1434	0.0642	0.1262	0.0959	0.1815	0.1157
%Improv.	15.34	14.96	14.21	15.07	12.28	12.04	13.42	12.89	7.63	7.15	6.82	7.09

- **RQ3:** How do the key designs in DGMAE contribute to its performance?
- **RQ4:** How do the key hyperparameters influence DGMAE’s performance?
- **RQ5:** In real cases, can DGMAE provide interpretation on its effectiveness in capturing global U-I relations and identifying effective U-I interactions?

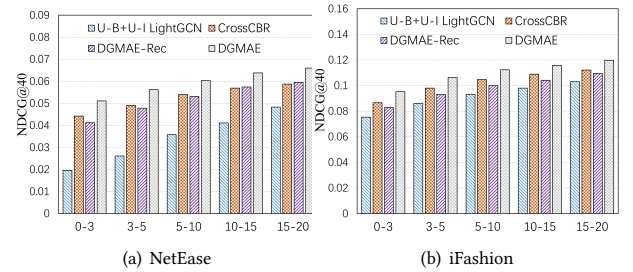
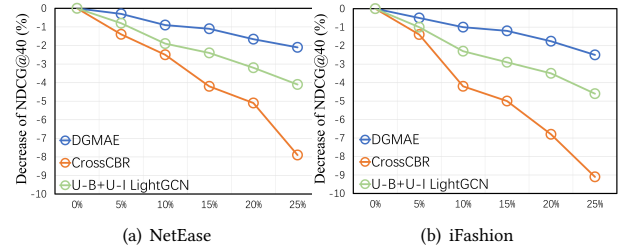
5.1 Experimental Settings

5.1.1 Datasets and Metrics. We evaluate DGMAE on the three datasets mentioned in Section 3.2, i.e., *Youshu*, *NetEase*, and *iFashion*. The dataset statistics are shown in Table 1. Each dataset is randomly split into training, validation, and testing sets with a ratio of 7:1:2. We employ Recall@K and NDCG@K as evaluation metrics to judge the ranking list performance, where $K \in \{20, 40\}$.

5.1.2 Baseline Methods. We compare DGMAE with the following state-of-the-art methods:

- **MFBPR [24]** is matrix factorization method based on a Bayesian Personalized Ranking learning framework.
- **DAM [4]** uses an attention network to aggregate item representations to bundle representations and model the U-I and U-B interactions in a multi-task manner.
- **BundleNet [6]** employs GNNs on the user-item-bundle tripartite graph and applies multi-task learning.
- **BGCN [3]** builds the bundle and item views and employs GNNs to separately learn user/bundle representations.
- **LightGCN [10]** learns user/bundle embeddings through linear propagation on the U-B graph. As the teacher model in DGMAE, it can provide an anchor of the performance.
- **MIDGN [34]** disentangles the user intents and item associations at a finer granularity and contrasts the user/bundle embeddings from different intent views.
- **CrossCBR [21]** follows the framework of BGCN and models the cooperative association between bundle and item views through a contrastive task.

5.1.3 Parameter Settings. For all the methods, we use the Xavier method [8] to initialize all trainable parameters and Adam [14] to optimize the parameters with a learning rate of 0.001. The embedding size d is set as 64. For our method, the hyperparameters p_l , p_u , λ , K , H are tuned from $\{0.1, 0.2, 0.3, 0.4\}$, $\{0.6, 0.7, 0.8, 0.9\}$, $\{0.001, 0.003, 0.01, 0.03\}$, $\{128, 256, 512, 1024\}$, $\{2, 4, 8\}$. The layer numbers of the encoders are all tuned from $\{1, 2, 3\}$ and the batch size

**Figure 5: Performance w.r.t U-B data sparsity degrees.****Figure 6: Performance w.r.t U-I data noise ratios.**

is set as 2048. For baseline methods, we quote the results of CrossCBR, BGCN, DAM, MFBPR, and BundleNet from [21] due to the same settings with us and implement the others by ourselves.

5.2 Performance Comparison (RQ1)

The comparison results of all methods are presented in Table 2. Based on the results, the following observations can be made:

- The models (BGCN and CrossCBR) separately modeling the U-B and U-I interactions consistently perform better than those (DAM and BundleNet) applying a holistic way. This result validates that simultaneously leveraging the two types of interactions cannot explicitly model the complex relations among users, items, and bundles.
- Compared to the bundle-specific methods, LightGCN is still a strong baseline and even outperforms BGCN due to its linear and light design. Meanwhile, our proposed DGMAE outperforms LightGCN, i.e., the teacher model by a large margin. This is because the knowledge of bundle view is extracted and injected into the student model which also benefits from the useful U-I interaction information. In addition, as analyzed in Section 3.2, the c -scores of *Youshu* and *NetEase* are quite low, indicating the inconsistency of U-I and U-B collaborative relations. And DGMAE has a

Table 3: Ablation Studies on key designs in DGMAE.

Dataset		Youshu		NetEase		iFashion	
Variants		Recall	NDCG	Recall	NDCG	Recall	NDCG
KD	-High	0.4154	0.2141	0.1374	0.0610	0.1753	0.1113
	-KD	0.3572	0.1849	0.1191	0.0533	0.1526	0.0971
	-Sum	0.3703	0.1941	0.1252	0.0556	0.1590	0.1015
	-CL	0.4106	0.2111	0.1364	0.0608	0.1728	0.1102
Mask	-Ada	0.4201	0.2159	0.1390	0.0621	0.1765	0.1122
	-Mask	0.4043	0.2071	0.1345	0.0601	0.1706	0.1086
	-Rec	0.3808	0.1963	0.1260	0.0565	0.1602	0.1019
Global	-VP	0.4006	0.2072	0.1321	0.0598	0.1744	0.1106
	-Glob	0.3510	0.1804	0.1156	0.0517	0.1661	0.1063
DGMAE		0.4323	0.2231	0.1434	0.0642	0.1815	0.1157

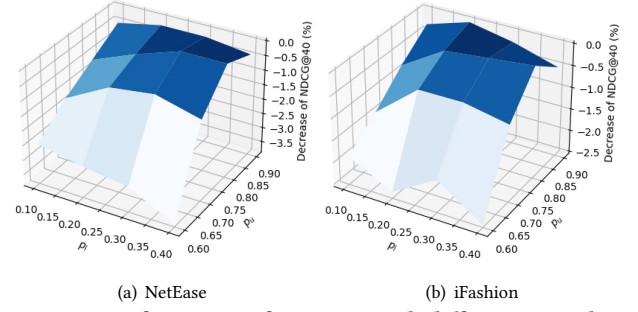
more significant improvement to LightGCN on these two datasets, demonstrating its effectiveness to alleviate this issue.

- Among all the methods, CrossCBR is the strongest baseline, implying the effectiveness of synthesizing the knowledge and modeling the cooperative association from two views. Our proposed DGMAE significantly outperforms CrossCBR, which can be attributed to: 1) DGMAE integrates the knowledge from two views in a more effective way through knowledge distillation and facilitates more fine-grained learning on user’s bundle preference. 2) The global transformer encoder enhances the representational power of DGMAE in depicting user’s bundle preference on the U-I graph. 3) The edge masked strategy and reconstruction target guide DGMAE to discover potential U-I interactions corresponding to user’s bundle preference.

5.3 Model Robustness Comparison (RQ2)

5.3.1 Performance w.r.t U-B Data Sparsity. In real scenarios, users always have extremely rare interactions with bundles than with items due to more expensive costs, leading to the sparsity issue of U-B interactions. To verify the robustness of DGMAE against the sparsity issue, we divide users into distinct groups according to their U-B interaction numbers in the training set (e.g., 10-15). We construct a variant of DGMAE by removing the masking operation and reconstruction loss (DGMAE-Rec) and compare DGMAE with DGMAE-Rec and two strong baselines CrossCBR, U-B+U-I LightGCN (mentioned in Section 3.2). From the results in Figure 5, we have the following observations: 1) As compared to the simple combination of two LightGCNs, CrossCBR achieves better performance, suggesting the effectiveness of incorporating associations between bundle and item views for data sparsity alleviation. 2) The edge masking and reconstruction is the key component in DGMAE to alleviate the sparsity issue, allowing the model to pay more attention to the effective U-I interactions of users with sparse U-B interactions. 3) DGMAE consistently achieves the best performance on different sparsity degrees, which validates the robustness of DGMAE in handling sparse data.

5.3.2 Performance w.r.t U-I Data Noise. As demonstrated by previous studies [31, 33], users may interact with “interest-irrelevant” items due to a variety of reasons. In bundle recommendation, the noisy U-I interactions may mislead the model to depict user’s bundle preference and further impair the recommendation performance. Therefore, to test the robustness of DGMAE to the noise issue, we substitute a certain ratio of U-I edges with randomly-sampled fake

**Figure 7: Performance of DGMAE with different p_l and p_u .**

edges and re-train the model on the corrupted U-I graphs. Specifically, we set the noisy ratio as {5%, 10%, 15%, 20%, 25%} and compare the performance degradation degree of DGMAE with CrossCBR and U-B+U-I LightGCN. We present the results in Figure 6 and have the following observations: 1) Compared to U-B+U-I LightGCN, CrossCBR is more sensitive to the data noise. We speculate the possible reason is that the CL task in CrossCBR transfers the noisy information from the item view to the bundle view, leading to larger performance degradation on both views. 2) DGMAE is robust to noisy data and presents the smallest performance degradation, which can be ascribed to two reasons. First, the edge masking and reconstruction task guide the model to identify the effective U-I interactions while ignoring the noisy ones. Second, the transformer encoder alleviates the noise effect of fake U-I interactions through global relation learning between users and items.

5.4 Ablation Studies (RQ3)

We conduct experiments on the three datasets to validate the effectiveness of key designs (i.e., the knowledge distillation object, the edge masking and reconstruction target, and the global transformer encoder) in DGMAE. Both the prominent components (e.g. the global encoder) and small modules (e.g. Vnode-level propagation) are ablated. The results of Recall@40 and NDCG@40 are reported in Table 3. We have the following observations:

- Removing the global transformer encoder (-Glob) severely hurts the performance, especially on Youshu and NetEase. This result suggests the necessity of capturing global collaborative relations and validates the expressive power improvements brought by the global encoder. In addition, we remove the Vnode-level propagation (-VP) in the global encoder and the result validates its contribution to global relation modeling through information exchange between user-side and item-side Vnodes.
- To explore the effect of edge masking and reconstruction, we substitute the adaptive masking strategy to a random one with probability $p \in \{0.5, 0.6, 0.7\}$ (-Ada), remove the masking operation (-Mask) and further remove the reconstruction loss (-Rec). From the results we can conclude that: 1) The variant without edge masking and reconstructing has obvious performance degradation, demonstrating their effectiveness in exploring the potential effective U-I interactions. 2) The masking operation improves the model performance by increasing the challenge of reconstructing task as well as the model’s ability in inferring unobserved ones. 3) The adaptive masking strategy outperforms the random one due to its focusing more on the users and items with sparse interactions and alleviating the sparsity issue.

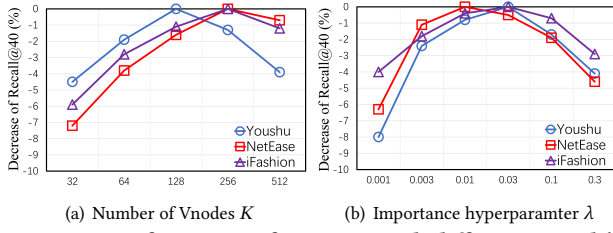


Figure 8: Performance of DGMAE with different K and λ .

- Replacing the KD loss with the BPR loss (-KD) leads to serious performance degradation due to its failure of combining the knowledge from the bundle view. Meanwhile, the effect of higher-order relation distillation can also be validated in the variant -High. To further compare the effects of different knowledge integration methods, we introduce two variants of DGMAE which simply optimize the BPR loss by summing the predicted results from two views (-Sum) and further adds the contrastive loss in CrossCBR (-CL). We find that DGMAE consistently outperforms its two variants in all cases, validating the superiority of our proposed KD loss in integrating the information from two views.

5.5 Hyperparameter Studies (RQ4)

In this section, we evaluate the effect of key hyperparameters (# of Vnodes K , hyperparameters p_l, p_u that control the range of masking probabilities and importance hyperparameter λ that balances the KD task and reconstruction task) in DGMAE and show the evaluation results in Figure 7 and 8.

- The values of K is examined in $\{32, 64, 128, 256, 512\}$. According to Figure 8, the performances of DGMAE first increase and then decrease on the three datasets, following a under- to over-fitting pattern. Generally, different K control the model’s capacity in global relation learning, and the best choice relies on how easy the over-fitting happens caused by the specific data size.
- We further investigate the impact of masking probability range by varying p_l from 0.1 to 0.4 and p_u from 0.6 to 0.9. As shown in Figure 7, the performance of DGMAE is relatively stable when p_u and p_l are set in a reasonable range (e.g. $p_l \in [0.2, 0.4]$ and $p_u \in [0.7, 0.9]$), demonstrating that DGMAE is robust to the perturbation of p_l and p_u . Overall, DGMAE prefers a higher masking probability on the effective U-I edges, which brings a higher challenge for reconstruction.
- We also study the impact of $\lambda \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3\}$ that weighs the contributions of KD task and reconstruction task. From Figure 8, the best values of λ on the three datasets all fall in the range of $[0.01, 0.03]$, which is consistent with our assumption. A proper weight of the reconstruction task can guide the model to identify effective U-I interactions and further predict potential U-B interactions at a finer granularity.

5.6 Case Study (RQ5)

In this section, we analyze the real data instances to investigate the effectiveness of DGMAE from two aspects: 1) Can DGMAE capture the global implicit U-I relations? 2) Can DGMAE identify potential effective U-I interactions that assist to predict users’ bundle preferences? To this end, we randomly select two users from NetEase dataset and present the original U-I interactions and the U-I relations learned by DGMAE and LightGCN for comparison.

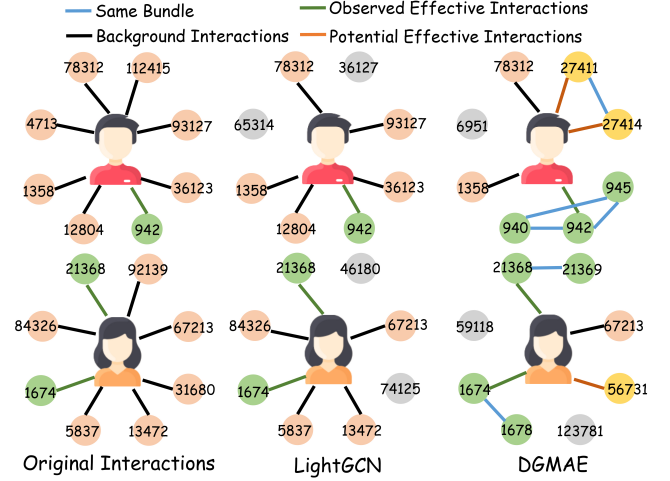


Figure 9: Case study on the U-I relations learned by DGMAE and LightGCN. Circles denote items interacted by the center user in the first column and the top items with the highest similarity scores to the center user in the last two columns. The numbers on the circles denote the item ids.

After training, we calculate the similarity scores $y_{u,v} = e_u^{S^T} e_v^S$ and present the items with top similarity scores. Meanwhile, we derive the observed and potential effective interactions from the training and test sets, respectively, to evaluate the learned relations. From the results in Figure 9, we have the following observations: 1) LightGCN tends to preserve the local U-I relations due to its iterative aggregation of neighbourhood information, whereas failing to differentiate the effective and background interactions. 2) Compared to LightGCN, DGMAE ignores the most of background interactions while preserving effective relations. In addition, DGMAE also captures the global implicit relations between users and items in the historically-interacted bundles. 3) DGMAE has the ability to infer the potential effective U-I interactions that provides guidance to the user’s bundle preferences. The results validate the effectiveness of DGMAE in recalibrating the U-I relations through global relation learning and effective relation identification.

6 CONCLUSION

In this paper, we conduct data analyses to provide guidance about how to leverage the U-I interactions in a more effective way for bundle recommendation. Based on the analyses, we propose a novel framework called DGMAE. Under the knowledge distillation framework, we distill relations among users and bundles from the bundle view to the item view for effective knowledge integration. In the student model, we enhance the GNN encoder with a global transformer encoder to increase the model’s expressive power. Meanwhile, an adaptive edge masking strategy and reconstruction target are designed to guide the model to identify the potential effective U-I interactions. Empirical results on benchmark datasets validate the superiority and the robustness of DGMAE compared to SOTA methods.

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